

Enhanced Learning for Special Education

Learning from home
with fun and ease



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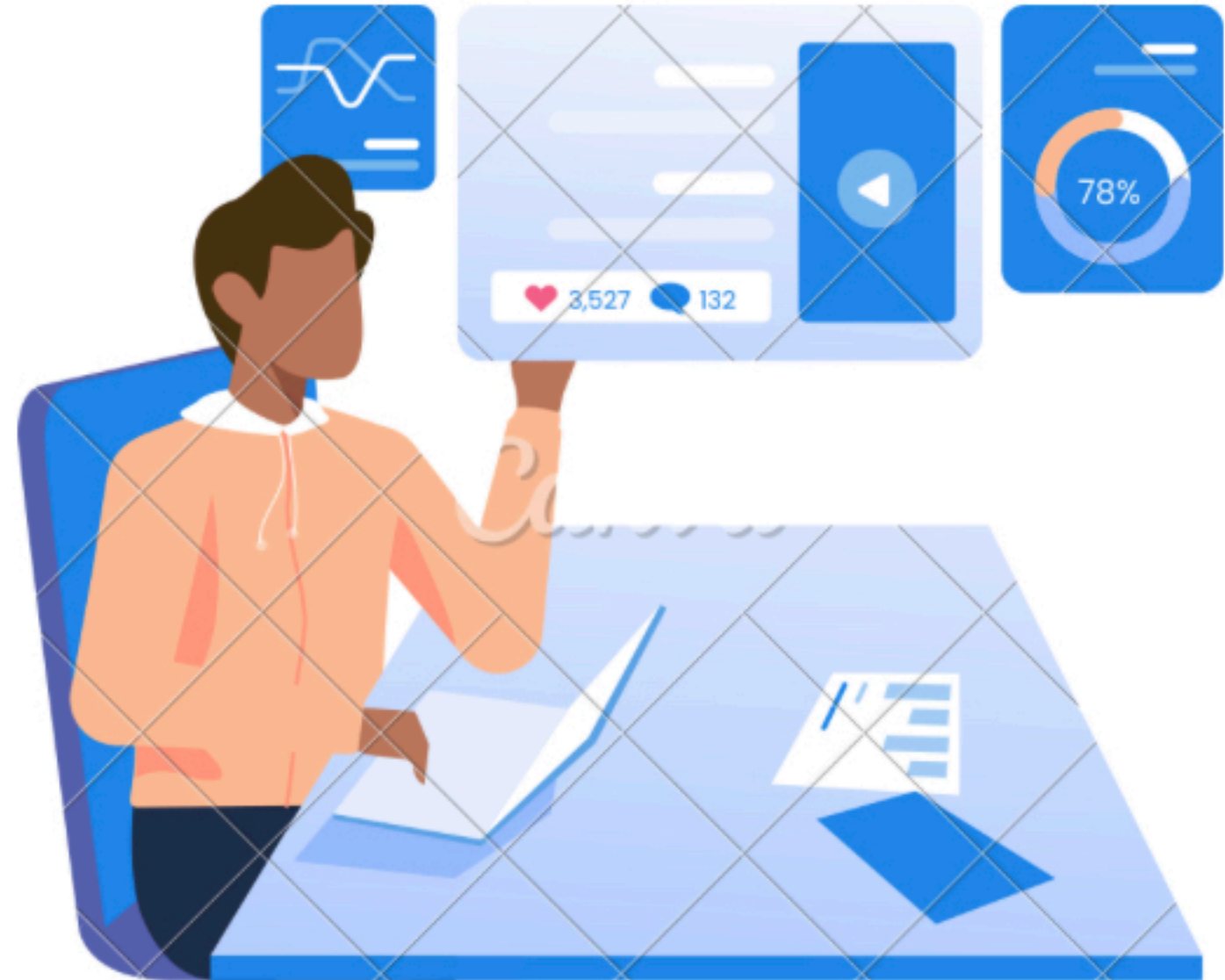


Project Idea Description

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The Enhanced Learning for Special Education project focuses on developing an interactive educational app to support students with learning difficulties, including dyslexia and ADHD.

- The app provides personalized learning paths, engaging multimedia content, progress tracking, and a resource library.



Targeted Users

Targeted Users:

1. Students with Learning Difficulties:

- Including those with dyslexia, ADHD, and other similar challenges.

2. Teachers:

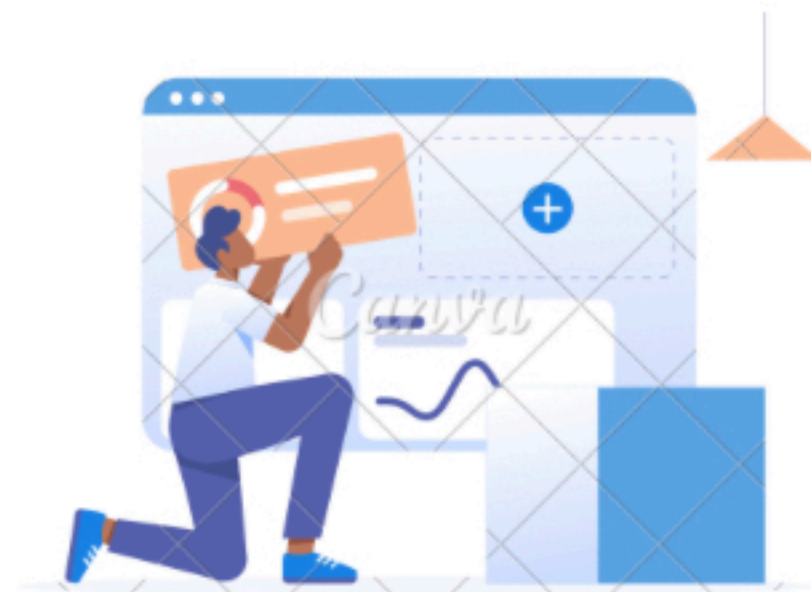
- Who require adaptive learning tools and resources to support diverse student needs.



Main Problems to Solve

Problems We Are Addressing:

1. Difficulty adapting traditional teaching methods to meet the needs of students with learning difficulties.
2. Lack of sufficient adaptive learning materials for both students and teachers.
3. Social and emotional impacts resulting from learning difficulties.



Overview of Interviews and Survey Questions

Data Collection Methods:

- We used surveys and interviews to gather feedback from students and teachers.
- Key survey/interview questions include:
 1. What academic field do you find challenging in terms of learning?
 2. Do you feel that you need additional time to understand new lessons?
 3. Do you find it difficult to concentrate during classes?



Raw Data Collected from Users

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	B
1	1. What academic field do you find challenging in terms of learning?
2	/2.
3	/2 Math and computer science
4	2/ Pure Mathematics: Fields like topology and algebraic geometry require d
5	2/ Math
6	2/ Reading
7	/1 H
8	/1 Arabic
9	/1 Math
10	/1 Java
11	/1 Arabic
12	/1 Math
13	/1 English
14	/1 Physics
15	/1 chemistry
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	D	
1	3. If yes, what difficulties have been diagnosed for you?	4. What teaching
2	- Other (please specify)	- Reading/Writin
3	- Mathematical difficulties	- Visual learning i
4	- Other (please specify)	- Reading/Writin
5	- Mathematical difficulties	- Visual learning i
6	- Writing difficulties	- Auditory learnir
7	- Writing difficulties	- Auditory learnir
8	- Writing difficulties	- Visual learning i
9	- Mathematical difficulties	- Kinesthetic lear
10	- Writing difficulties	- Kinesthetic lear
11	- Reading difficulties	- Reading/Writin
12	- Mathematical difficulties	- Kinesthetic lear
13	- Writing difficulties	- Reading/Writin
14	- Mathematical difficulties	- Kinesthetic lear
15	- Mathematical difficulties	- Visual learning i

Responses show students find Math, Arabic, and Writing the most challenging.

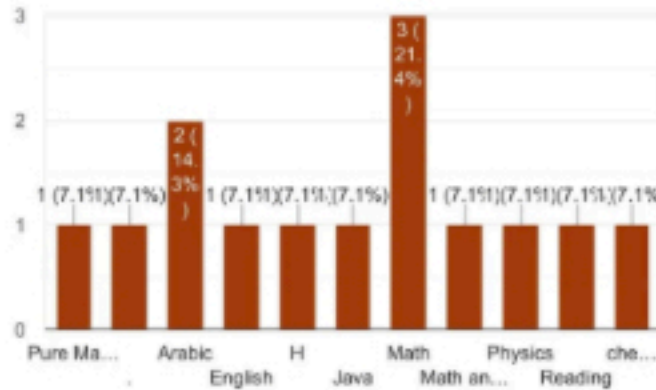
Diagnosed difficulties include: Mathematical (7), Writing (5), Reading (1), Other (2).

Preferred learning methods: Kinesthetic (4), Visual (4), Reading/Writing (3), Auditory (2).



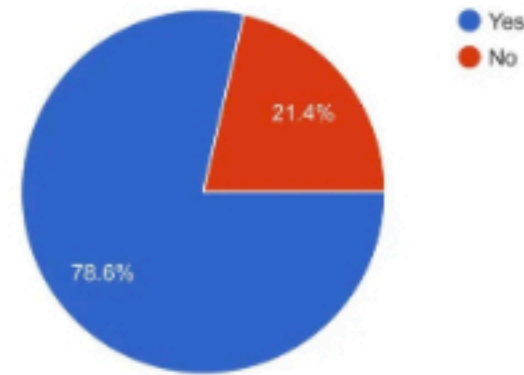
Data Visualization

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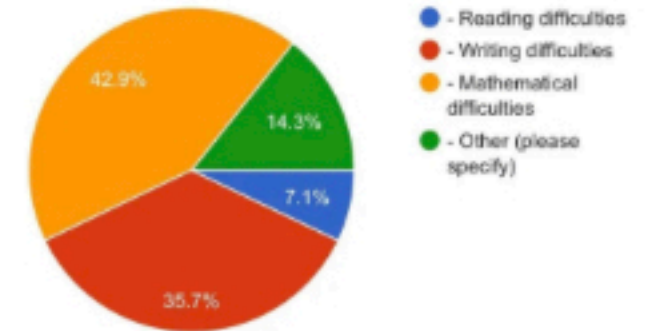
Bar Chart – Most Challenging Subjects

The bar chart shows which subjects respondents find most challenging. Mathematics stands out as the top challenge (21.4% of responses), with Arabic as the next-highest (14.3%). All other subjects (English, Physics, Chemistry, etc.) were chosen by only one respondent each (7.1%).



Pie Chart – Formal Evaluation

The pie chart shows that 78.6% of respondents answered "Yes" (blue) and 21.4% answered "No" (red) to having been formally evaluated for learning difficulties. In other words, 11 out of 14 respondents have undergone such an assessment, while 3 have not. Thus, roughly 79% of respondents have been formally assessed for learning difficulties.



Pie Chart – Diagnosed Difficulties

The final pie chart breaks down the types of diagnosed learning difficulties. It shows 42.9% of respondents reported mathematical difficulties, 35.7% reported writing difficulties, 14.3% reported other difficulties, and 7.1% reported reading difficulties. Mathematical difficulties (42.9%) are the most commonly diagnosed issue among respondents.

Key Needs and Insights

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Personalized Learning:

- Students need tailored learning paths to suit their individual challenges.

Interactive Content:

- Students prefer engaging multimedia resources like videos and quizzes.

Continuous Feedback:

- Students need constant feedback to stay motivated and improve.

Teacher Support:

- Teachers need resources and tools to customize the learning experience.

Collaboration Tools:

- Communication and collaboration tools are essential for teacher-student interaction.



Tasks to Meet User Needs

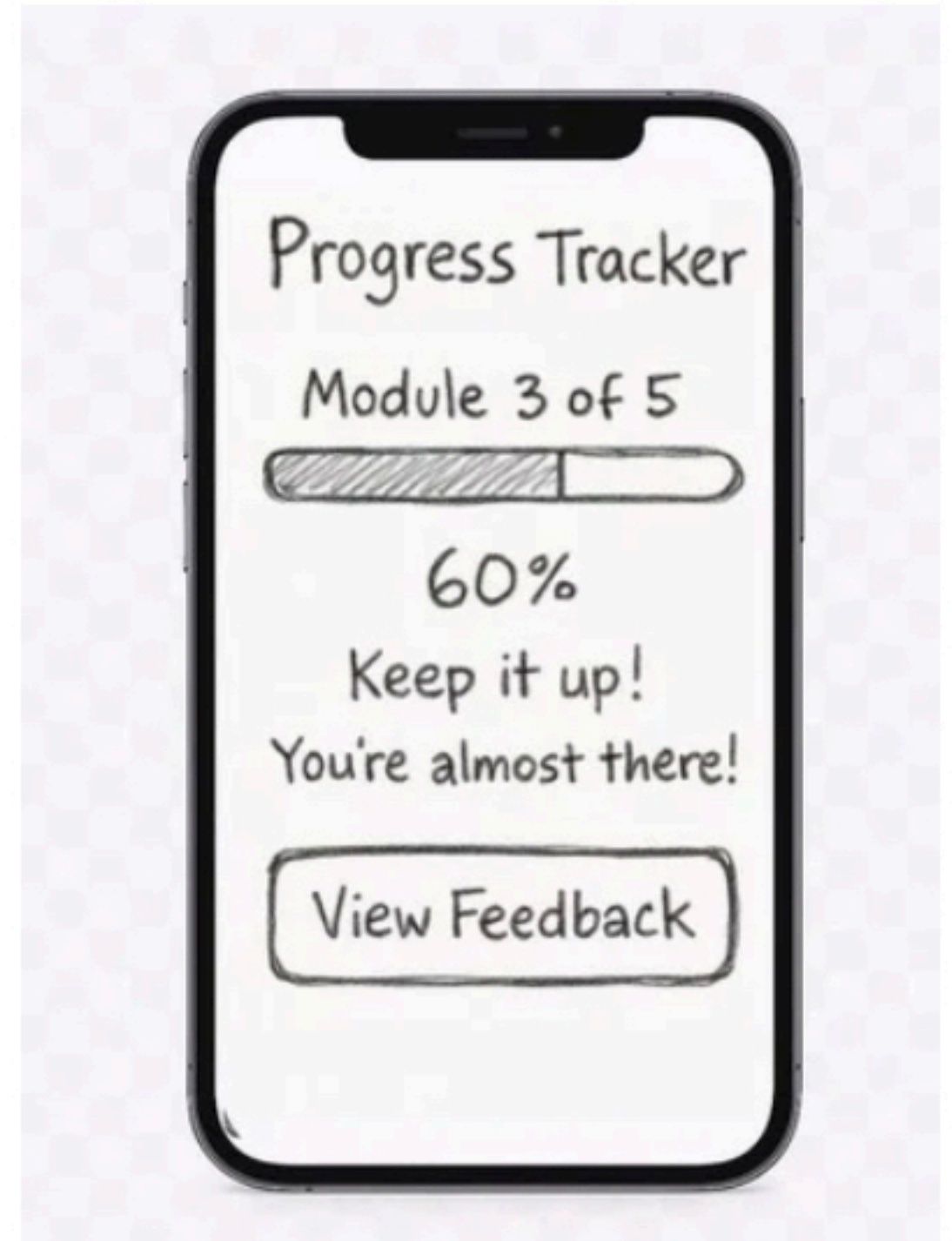
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1. Develop personalized learning paths based on students' needs.
2. Integrate multimedia content (videos, animations, quizzes) for higher engagement.
3. Implement progress tracking and feedback features.
4. Build a resource library for teachers with adaptable content.
5. Add social communication tools for collaboration between students and teachers.

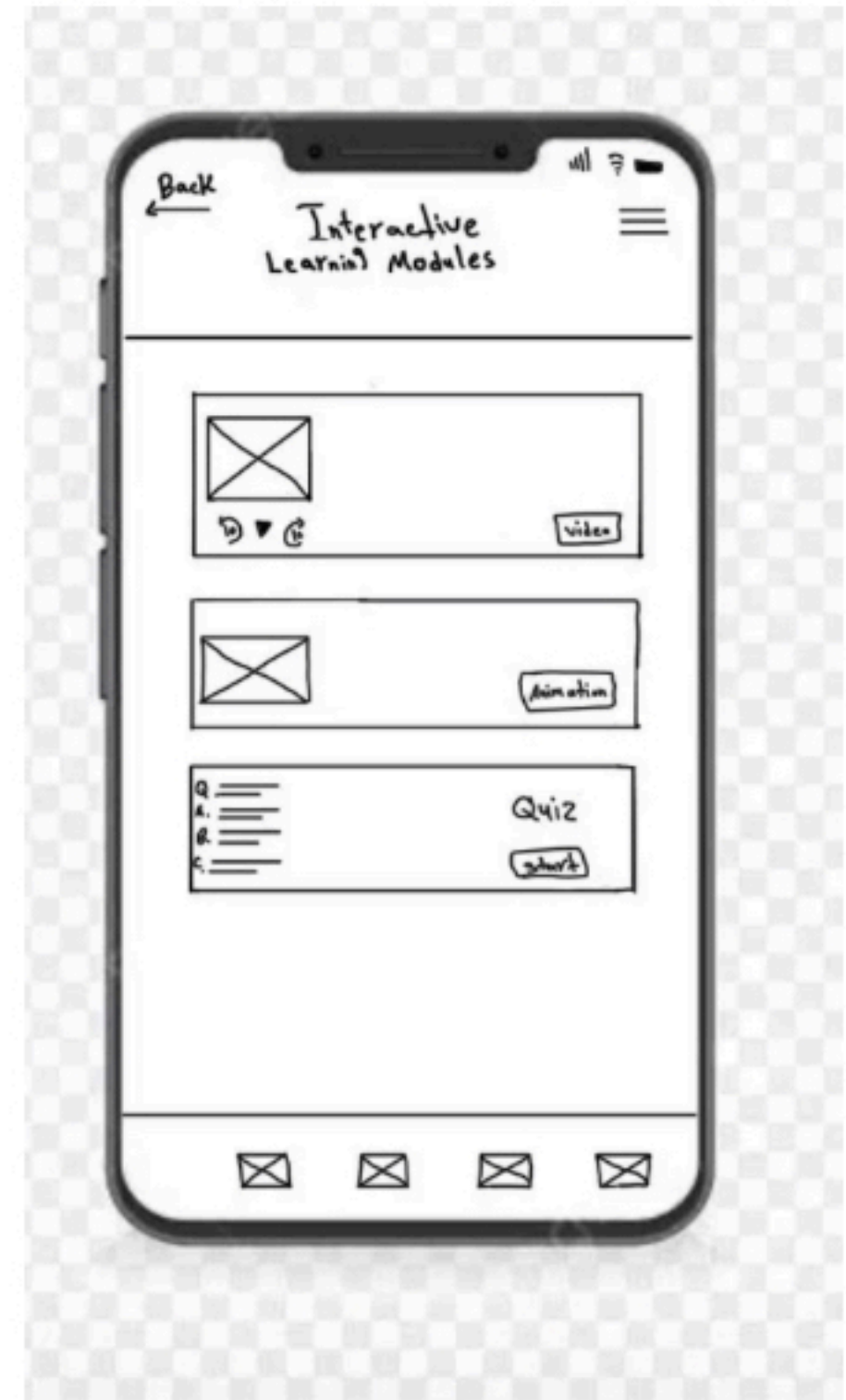


The prototype

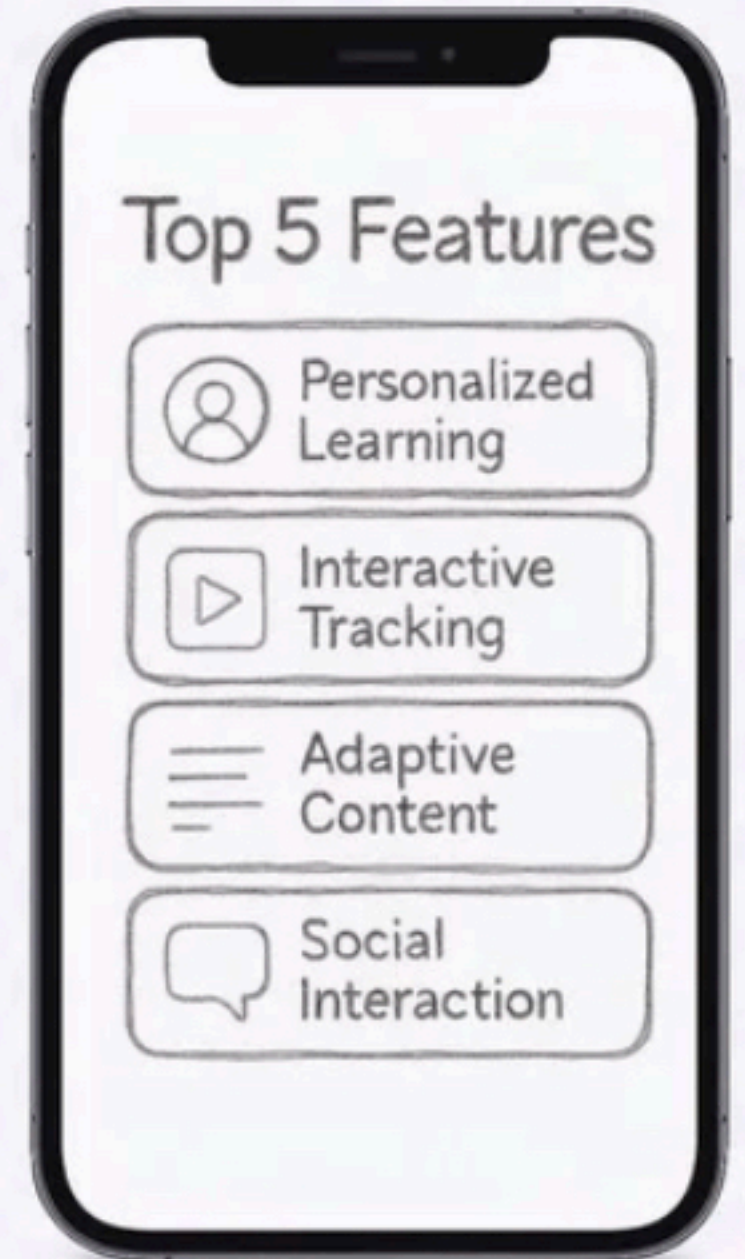
1. Progress Indicator: Shows the user is on Module 3 of 5.
2. Completion Percentage: Displays 60% completion.
3. Encouragement Message: Motivates the user to keep going.
4. Feedback Button: Provides access to performance feedback.



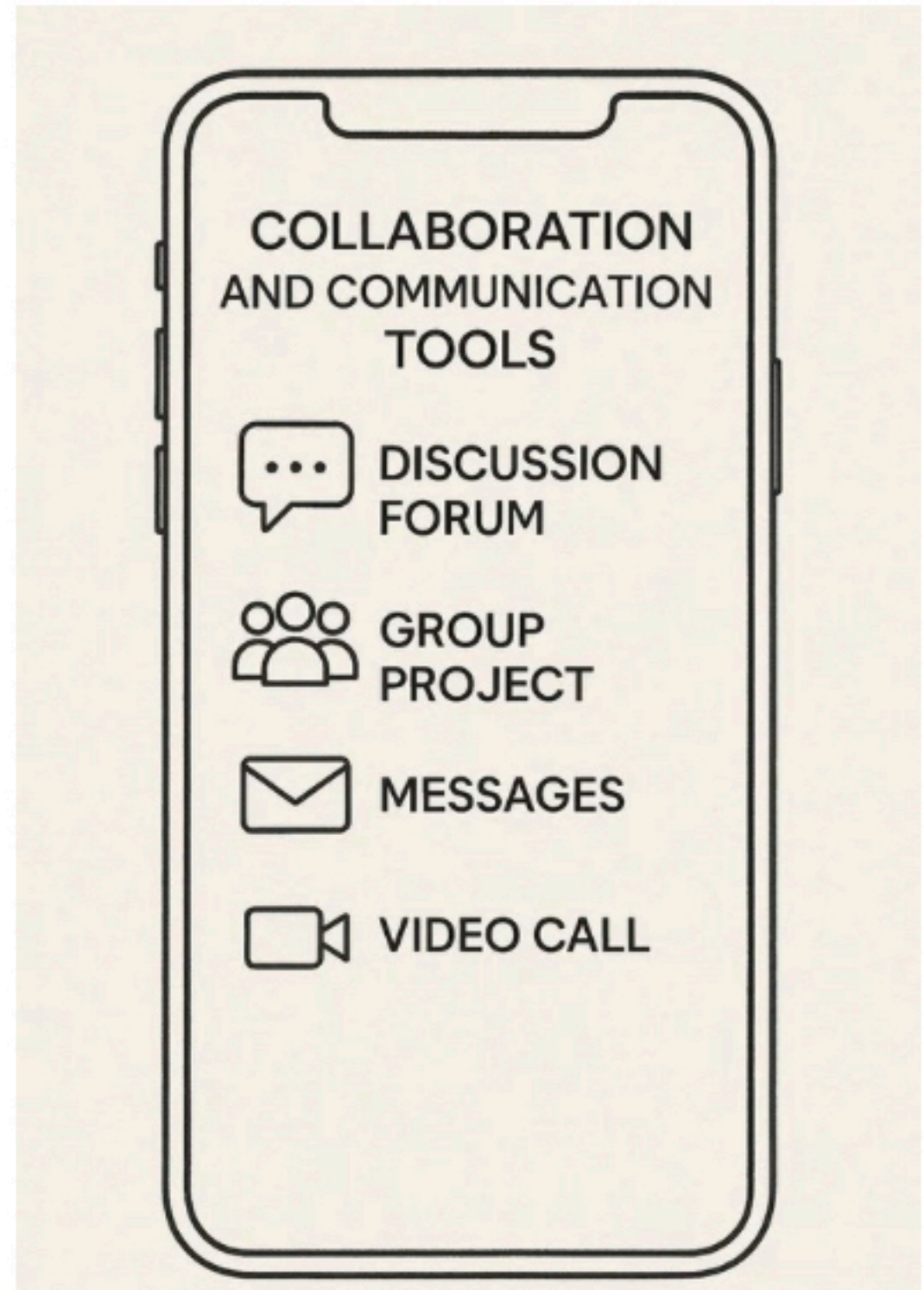
1. Navigation Bar: "Back" button to return.
2. Module Titles: Lists available learning modules.
3. Content Previews: Placeholders for videos and animations.
4. Quiz Section: Area for quizzes.



1. Header: Lists top features.
2. Personalized Learning: Customizes content.
3. Interactive Tracking: Monitors progress.
4. Adaptive Content: Adjusts based on performance.
5. Social Interaction: Enables user communication.



1. Header: Collaboration tools.
2. Discussion Forum: User discussions.
3. Group Project: Collaborative work.
4. Messages: Private communication.
5. Video Call: Real-time conversations.



New Design Features

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How the New Design Meets User Requirements:

1. Personalization:

- Adaptive learning paths based on individual needs.

2. Engagement:

- Interactive multimedia content to keep students interested.

3. Feedback:

- Continuous progress tracking and feedback to improve learning outcomes.

4. Teacher Resources:

- A library to support teachers with materials and lesson plans.

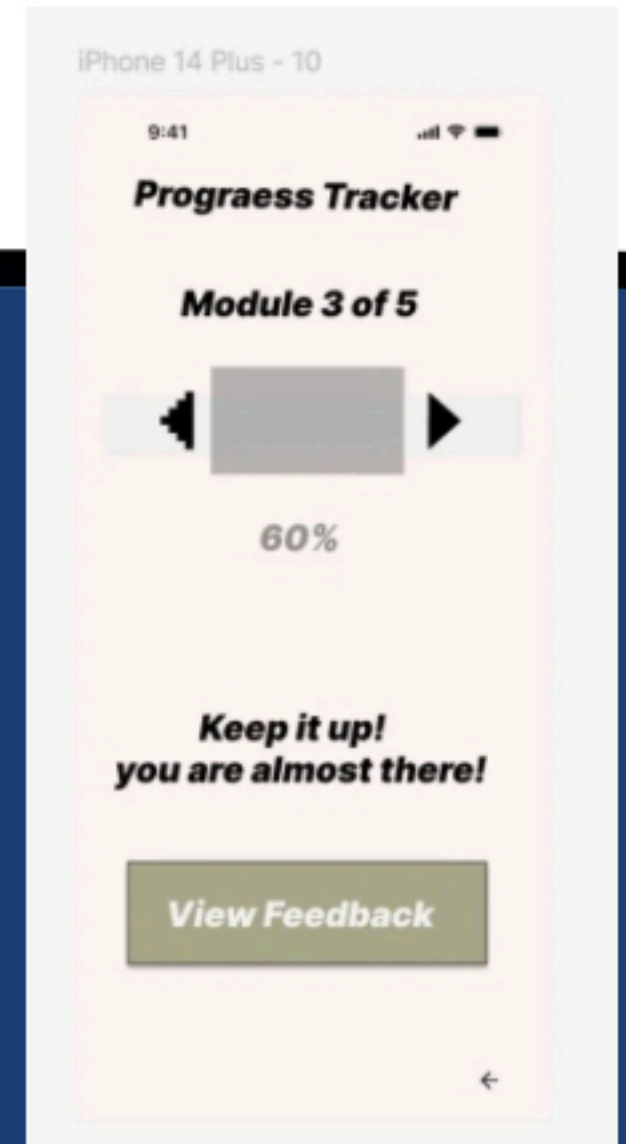
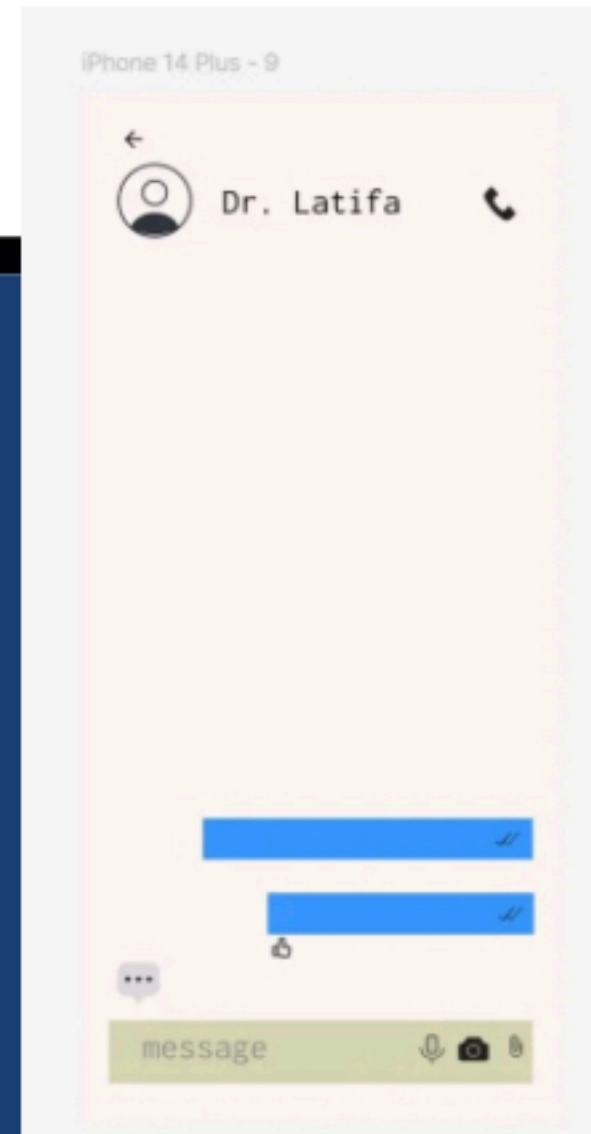
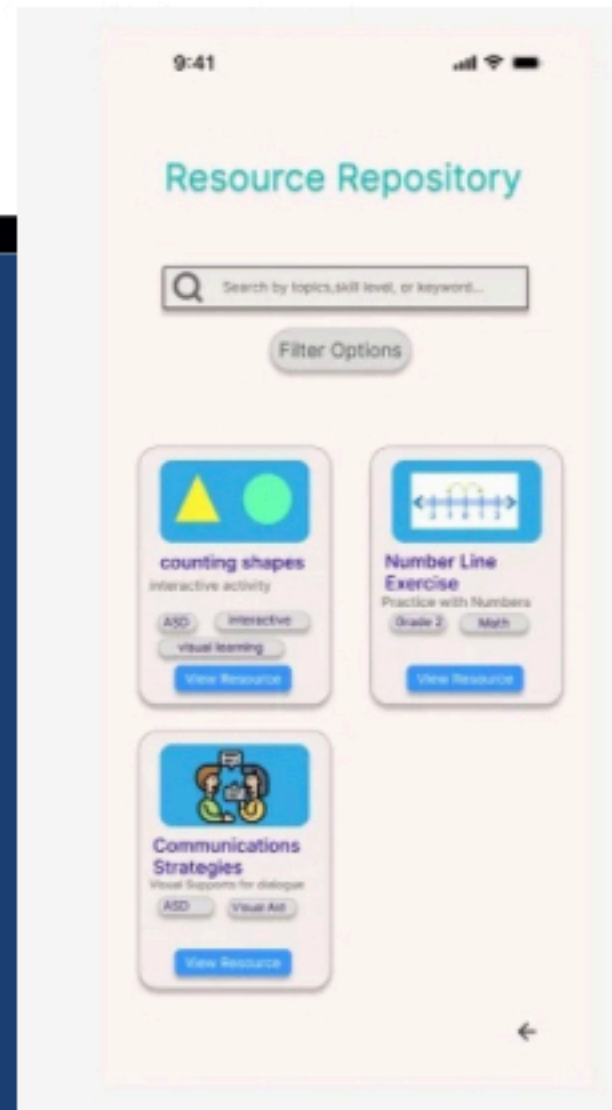
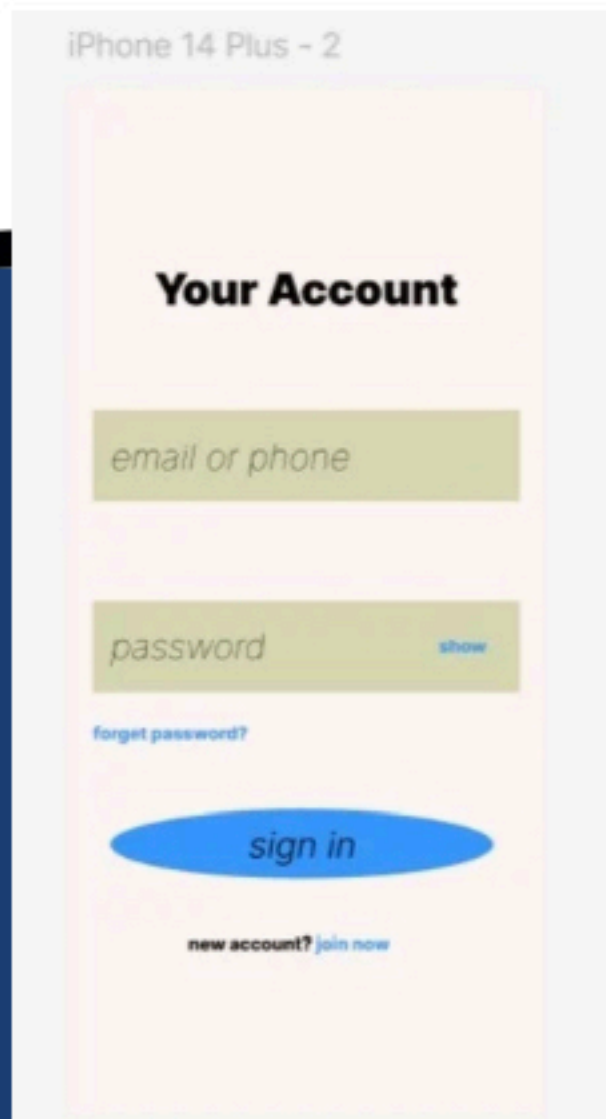
5. Collaboration:

- Tools to enhance communication and collaboration between teachers and students.



High-fidelity Prototype

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Challenges and Further Improvements

Challenges Faced:

1. Ensuring the app meets the needs of a diverse group of learners.
2. Balancing simplicity with the functionality of the app.
3. Adapting multimedia content to be accessible to all students.

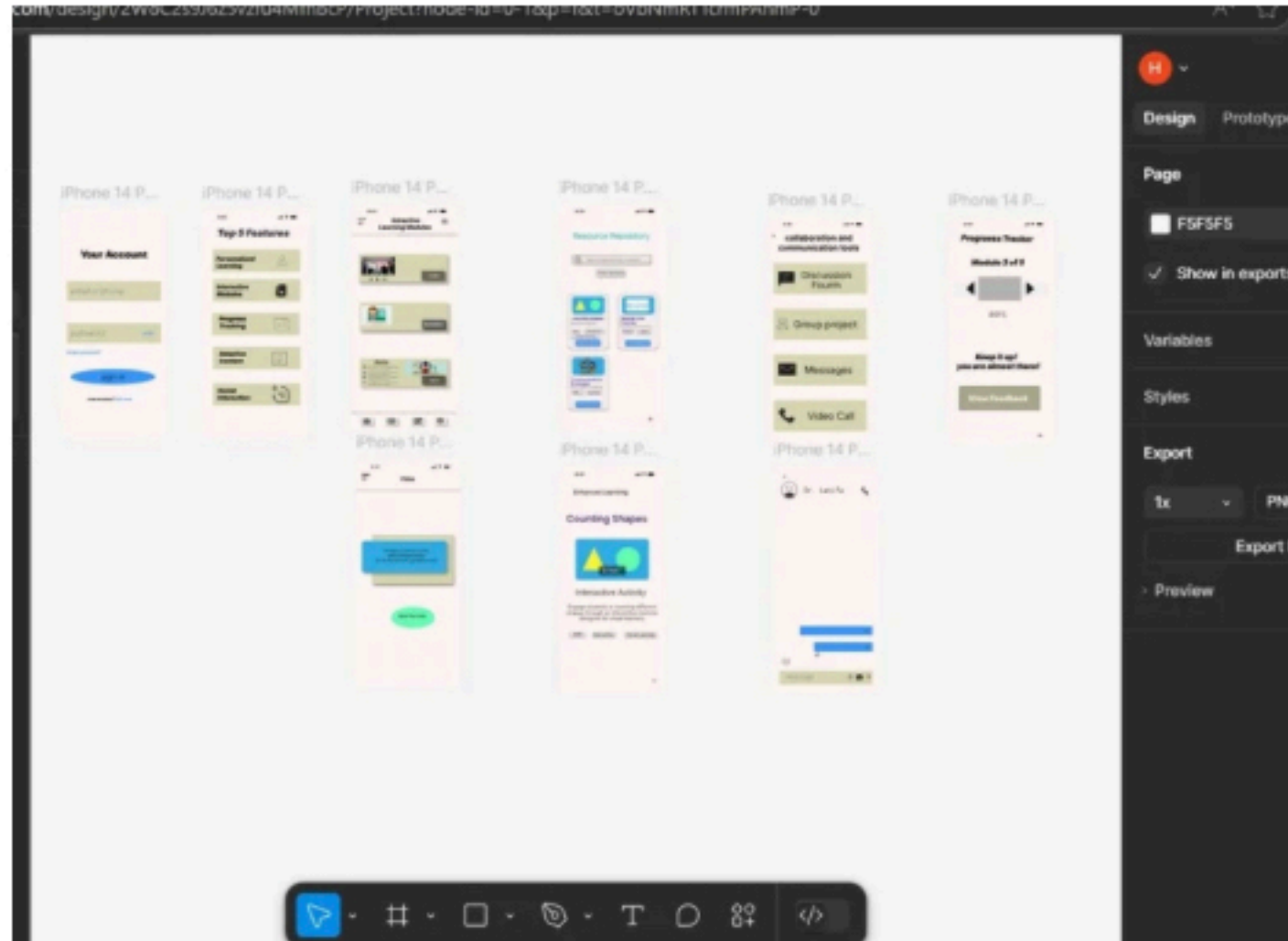
Further Improvements:

1. Continuously update content based on user feedback.
2. Expand features for social learning and peer interaction.
3. Integrate accessibility features for students with various disabilities.



Demo of high-fidelity prototype (figma)

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This demo showcases how users interact with the app using Figma

Thank you !!

"Thank you for your attention! Any questions?"

